

Geomathematics

Geomathematics (also: **mathematical geosciences**, **mathematical geology**, **mathematical geophysics**) is the application of [mathematical](#) methods to solve problems in [geosciences](#), including [geology](#) and [geophysics](#), and particularly [geodynamics](#) and [seismology](#).

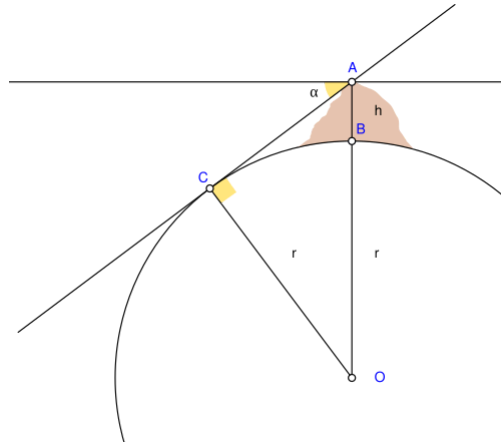


Diagram of [Al-Biruni's](#) method, involving [geometry](#) and [surveying](#), for determining Earth's [radius](#) using the height of a [mountain](#) and the dip angle to the [horizon](#). This early application of [trigonometry](#) and [algebra](#) to [geophysical](#) measurement is a foundational example of **geomathematics**.



Applications

Geophysical fluid dynamics

[Geophysical fluid dynamics](#) develops the theory of [fluid dynamics](#) for the atmosphere, the ocean, and Earth's interior.^[1] Applications include geodynamics and the theory of the [geodynamo](#).

Geophysical inverse theory

Geophysical [inverse theory](#) is concerned with analyzing geophysical data to get model parameters.^{[2][3]} It is concerned with the question: What can be known about the Earth's interior from measurements on the surface? Generally there are limits on what can be known even in the ideal limit of exact data.^[4]

The goal of inverse theory is to determine the spatial distribution of some variable (for example, density or seismic wave velocity). The distribution determines the values of an observable at the surface (for example, gravitational acceleration for density). There must be a *forward model* predicting the surface observations given the distribution of this variable.

Applications include [geomagnetism](#), [magnetotellurics](#) and seismology.

Fractals and complexity

Many geophysical data sets have spectra that follow a [power law](#), meaning that the frequency of an observed magnitude varies as some power of the magnitude. An example is the distribution of [earthquake](#) magnitudes; small earthquakes are far more common than large earthquakes. This is often an indicator that the data sets have an underlying [fractal](#) geometry. Fractal sets have a number of common features, including structure at many scales, irregularity, and [self-similarity](#) (they can be split into parts that look much like the whole). The manner in which these sets can be divided determine the [Hausdorff dimension](#) of the set, which is generally different from the more familiar [topological dimension](#). Fractal phenomena are associated with [chaos](#), [self-organized criticality](#) and [turbulence](#).^[5] Fractal Models in the Earth Sciences by [Gabor Korvin](#) was one of the earlier books on the application of [Fractals](#) in the [Earth Sciences](#).^[6]

Data assimilation

[Data assimilation](#) combines numerical models of geophysical systems with observations that may be irregular in space and time. Many of the applications involve geophysical fluid dynamic. Fluid dynamic models are governed by a set of [partial differential equations](#). For these equations to make good predictions, accurate initial conditions are needed. However, often the initial conditions are not very well known. Data assimilation methods allow the models to incorporate later observations to improve the initial conditions. Data assimilation plays an increasingly important role in [weather forecasting](#).^[7]

Geophysical statistics

Some statistical problems come under the heading of mathematical geophysics, including [model validation](#) and quantifying uncertainty.

Terrestrial Tomography

An important research area that utilises inverse methods is [seismic tomography](#), a technique for imaging the subsurface of the Earth using [seismic waves](#). Traditionally seismic waves produced by [earthquakes](#) or anthropogenic seismic sources (e.g., explosives, marine air guns) were used.

Crystallography

[Crystallography](#) is one of the traditional areas of [geology](#) that use [mathematics](#).

Crystallographers make use of [linear algebra](#) by using the [Metrical Matrix](#). The [Metrical Matrix](#) uses the basis vectors of the [unit cell](#) dimensions to find the volume of a unit cell, d-spacings, the angle between two planes, the angle between atoms, and the bond length.^[8] Miller's Index is also helpful in the application of the [Metrical Matrix](#). [Brag's equation](#) is also useful when using an [electron microscope](#) to be able to show relationship between light diffraction angles, wavelength, and the d-spacings within a sample.^[8]

Geophysics

Geophysics is one of the most **math** heavy disciplines of **Earth Science**. There are many applications which include **gravity**, **magnetic**, **seismic**, **electric**, **electromagnetic**, **resistivity**, radioactivity, induced polarization, and **well logging**.^[9] Gravity and magnetic methods share similar characteristics because they're measuring small changes in the gravitational field based on the density of the rocks in that area.^[9] While similar **gravity fields** tend to be more uniform and smooth compared to **magnetic fields**. Gravity is used often for **oil exploration** and seismic can also be used, but it is often significantly more expensive.^[9] Seismic is used more than most geophysics techniques because of its ability to penetrate, its resolution, and its accuracy.

Geomorphology

Many applications of **mathematics** in **geomorphology** are related to water. In the **soil** aspect things like **Darcy's law**, **Stokes' law**, and **porosity** are used.

- **Darcy's law** is used when one has a saturated soil that is uniform to describe how **fluid flows** through that medium.^[10] This type of work would fall under **hydrogeology**.
- **Stokes' law** measures how quickly different sized particles will settle out of a fluid.^[10] This is used when doing **pipette analysis** of soils to find the percentage sand vs silt vs clay.^[11] A potential error is it assumes perfectly spherical particles which don't exist.
- **Stream power** is used to find the ability of a river to **incise** into the **river bed**. This is applicable to see where a river is likely to fail and change course or when looking at the damage of losing stream sediments on a river system (like downstream of a dam).
- **Differential equations** can be used in multiple areas of **geomorphology** including: The **exponential growth equation**, distribution of sedimentary rocks, **diffusion of gas** through rocks, and **crenulation** cleavages.^[12]



Glaciology

Mathematics in **Glaciology** consists of theoretical, experimental, and modeling. It usually covers **glaciers**, **sea ice**, **waterflow**, and the land under the glacier.

Polycrystalline ice deforms slower than single crystalline ice, due to the stress being on the basal planes that are already blocked by other ice crystals.^[13] It can be **mathematically modeled** with **Hooke's law** to show the elastic characteristics while using **Lamé constants**.^[13] Generally the ice has its linear **elasticity** constants averaged over one dimension of space to simplify the equations while still maintaining accuracy.^[13]

Viscoelastic polycrystalline ice is considered to have low amounts of **stress** usually below one **bar**.^[13] This type of ice system is where one would test for **creep** or **vibrations** from the **tension** on the ice. One of the more important equations to this area of study is called the relaxation

function.^[13] Where it's a [stress-strain](#) relationship independent of time.^[13] This area is usually applied to transportation or building onto floating ice.^[13]

Shallow-Ice approximation is useful for [glaciers](#) that have variable thickness, with a small amount of stress and variable velocity.^[13] One of the main goals of the mathematical work is to be able to predict the stress and velocity. Which can be affected by changes in the properties of the ice and temperature. This is an area in which the basal shear-stress formula can be used.^[13]

Academic journals

- [International Journal on Geomathematics](#)
- [Mathematical Geosciences](#)

See also

- [Geocomputation](#)
- [Geoinformatics](#)
- [International Association for Mathematical Geosciences](#) (IAMG)



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Further reading

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